CAPSTONE 1 · AI PRODUCT ENGINEERING

Beyond Arrival

A white-label companion that keeps supporting a relocated family after the relocation provider’s contract ends.

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THE PROBLEM · RELOCATION PROVIDERS

Relocation providers stop at week six. The hard months come after.

When a company sends an employee abroad, it pays a relocation provider to move the family: flat, visa paperwork, registration, schools.

Weeks 0 to 6

The provider is on contract and on call.

Months 3 to 12

Kita and school, health insurance, the partner’s work rights, a job search, loneliness. Nobody is on contract any more.

THE SYSTEM

Safety is decided in code. The model writes.

1 · CODE

Remove personal data

IBAN, phone, tax ID never reach a model

2 · MODEL

Crisis check

Qwen3.8 27B on Groq: needs a person?

3 · CODE

Advice limits

No individual legal, visa or tax advice

4 · SEARCH

Find 6 cards

bge-m3 + pgvector, 64 chunks of 20 cards

5 · MODEL

Write the answer

gpt-oss 120B, only from the cards

6 · CODE

Check the answer

Numbers and claims must be in the cards

Steps 2 and 3 can stop a message before any card is read. Step 6 removes a sentence the cards do not back, instead of the whole answer.

Dark boxes: plain code or search, no model judgement. Light boxes: a hosted model.

LIVE DEMO · 4 MINUTES

Five messages, five behaviours

What it shows	What I type
An answer from the cards	How do I register our address?
It hears the feeling	My husband loves his new job here and I just feel lost. Is that normal?
Personal data removed	Our Kita wants a direct debit. My IBAN is DE89 3704 0044 0532 0130 00 …
A person, not a bot	My wife hasn’t left the flat for a week and barely talks to me.
Where advice stops	Will my residence permit be approved if I change jobs?

Runs on this laptop: local database and search, hosted models. The IBAN is the standard published example number.

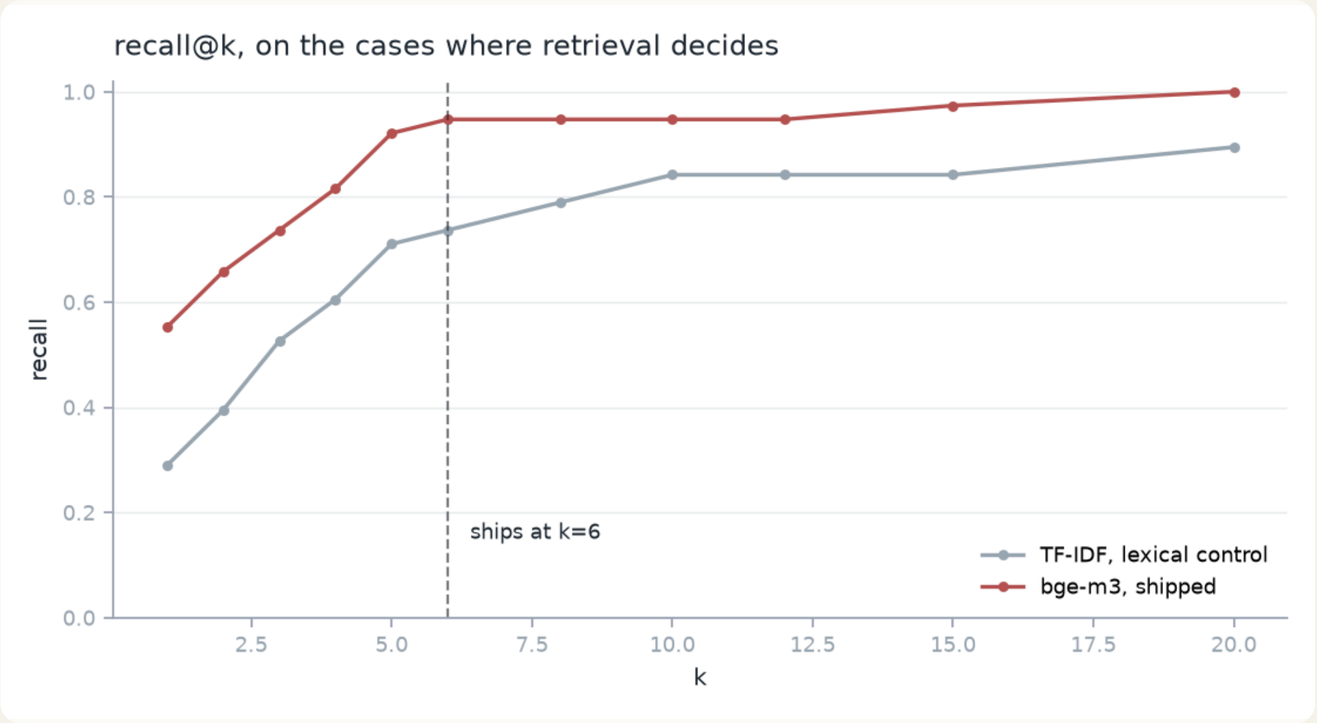
EDA · RETRIEVAL

The curve chose six cards

0.947

of the cases that need a card find it in the top 6. Flat from 6 to 12, so 6 ships.

+0.21 over word matching (TF-IDF) at k = 6: meaning-based search earns its cost.



MODELS AGAINST A BASELINE

Tier 2 ships: best at relevance and next steps

Setup (same 115 messages)	Grounded	Relevant	Next step	Answered
Baseline 0: best card, pasted	0.92	0.37	–	42
Tier 1: llama3.1 8B, local	0.71	0.56	0.34	60
Tier 2: gpt-oss 120B, hosted	0.86	**0.81**	**0.79**	61
Tier 3: nemotron-3-ultra, hosted	0.88	0.80	0.70	60
Tier 4: gemma4 31B, hosted	0.80	0.72	0.74	61
Cohen's κ: judge vs my labels	*0.13*	*0.16*	*n/a*	*counted*

Scores come from an LLM judge (qwen3 14B) that agrees with my labels barely above chance; so read them as direction. The best judge today, gemma4 31B, reaches κ 0.00 / 0.21 / 0.38.

Cohen's κ = agreement beyond chance: 0 is a coin flip, 1 is perfect, 0.6 is the pass mark

IMPROVING IT: ONE CHANGE AT A TIME

Code and content changes worked. Prompt rules mostly did not.

Step	Change	Kind	Result
E0	Same setup run twice	control	shows what is chance
E1, E2	Drop only unbacked sentences; numbers by value	code	kept
E3	Clarifying replies see no cards	code	kept
E4	Five new prompt rules	prompt	2 kept, 3 did nothing
E6	Two missing facts written into the cards	content	both target cases fixed
E5, E7	Sharper handoff rules; permit loss goes to a person	policy	handoff recall 0.42 → 0.92

One change per run, after a control run that shows how much is chance

SAFETY · HANDING OVER TO A PERSON

11 of 12 posts that need a person now reach one

Recall: needed a person, got one

0.42 → 0.92

24 held-out forum posts

Precision: handed over, and needed it

0.63 → 0.85

same posts, same check

Honest caveat: I tuned the rules on these posts, so 0.92 is a fitted number. The price showed in testing this week: mild messages such as “six months in, it still doesn’t feel like home” also go to a person.

Crisis check: qwen3.8 27B on Groq · on the development set 0.66 → 0.97, also fitted

LIMITS AND NEXT STEPS

What I would build next, in this order

1 · Who is speaking about whom

The family profile never reaches the model yet. 20 test cases are written: same words, different family, different right answer.

2 · Memory between messages

A guiding question only helps if the next message knows it was asked.

3 · A fresh, unfitted test set

New posts labelled against the written handoff definition, to measure recall and the over-referral honestly.

4 · A judge I can trust

More “no” examples in my labels, not a bigger judge: with 29 of 30 grounded, Cohen’s κ stays at chance.

THANK YOU

Try it yourself

Measure first. Put safety in code. Fix knowledge in the cards.

Questions?

Live companion

beyond-arrival-deploy
.onrender.com

Access code: in the talk

Code and evaluations

github.com/sean-mas/
beyond-arrival-capstone